Terraform Tasks - 02

- Execute all the **templates shown in the video**.
- **Lifecycle Rules:**

main.tf > resource "random_pet" "my_pet"

```
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Miss"
  separator = "."
  length    = 1

  lifecycle {
    create_before_destroy = true
  }
}
```

-

```
random_pet.my_pet: Creating...
random_pet.my_pet: Creation complete after 0s [id=Miss.mammal]
random_pet.my_pet (deposed object b8ac9d84): Destroying... [id=Mr.gibbon]
random_pet.my_pet: Destruction complete after 0s

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

main.tf > resource "random_pet" "my_pet"

```
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Miss"
  separator = "."
  length    = 1

  lifecycle {
    prevent_destroy = true
  }
}
```

-

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```
Plan: 0 to add, 0 to change, 2 to destroy.

Error: Instance cannot be destroyed

  on main.tf line 6:
   6: resource "random_pet" "my_pet" {

Resource random_pet.my_pet has lifecycle.prevent_destroy set, but the plan calls for this resource to be destroyed. To avoi
disable lifecycle.prevent_destroy or reduce the scope of the plan using the -target option.

PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

main.tf > resource "random_pet" "my_pet" > prefix

```
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = 1

  lifecycle {
    ignore_changes = [prefix]
  }
}
```

-

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform apply
random_pet.my_pet: Refreshing state... [id=Miss.mammal]
local_file.my_pet: Refreshing state... [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

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- main.tf > resource "local_file" "my_file"

```tf
resource "random_pet" "my_pet" {
  prefix = "Miss"
  separator = "."
  length = 21
}

resource "local_file" "my_file" {
  filename = "pets.txt"
  content  = random_pet.my_pet.id

  lifecycle {
    replace_triggered_by = [
      random_pet.my_pet
    ]
  }
}
```

```
}

Plan: 2 to add, 0 to change, 2 to destroy.
local_file.my_file: Destroying... [id=9c444f11ce30bdd29bae1b1ff09d3396d2db4163]
local_file.my_file: Destruction complete after 0s
random_pet.my_pet: Destroying... [id=Mr.hamster]
random_pet.my_pet: Destruction complete after 0s
random_pet.my_pet: Creating...
random_pet.my_pet: Creation complete after 0s [id=Miss.thoroughly.monthly.verbally.rapidly.nicely.gladly.remarkably.indirec
inly.locally.abnormally.freely.naturally.wildly.exact.turkey]
local_file.my_file: Creating...
local_file.my_file: Creation complete after 0s [id=6a7398087231f30bf18cf7144acd27bb2bf8d531]

Apply complete! Resources: 2 added, 0 changed, 2 destroyed.
```

- **Variables:**

main.tf > resource "random_pet" "my_pet"

```tf
resource "local_file" "my_pet" {
  filename = var.filename
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = 1
}
```

variables.tf > variable "filename"

```tf
variable "filename" {
  default = "Clubs.txt"
  type = string
  description = "filename"
}
```

-

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main.tf | Clubs.txt | variables.tf

Clubs.txt

```
I love pets!
```

main.tf > resource "random_pet" "my_pet"

```
resource "local_file" "my_pet" {
  filename = var.filename
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = var.length
}
```

variables.tf > variable "length"

```
variable "filename" {
  default = "Clubs.txt"
  type = string
  description = "filename"
}
variable "length" {
  default = "2"
  type = number
  description = "Length of random pet"
}
```

-

```
      ~ length     = 1 -> 2 # forces replacement
        # (2 unchanged attributes hidden)
    }

Plan: 1 to add, 0 to change, 1 to destroy.
random_pet.my_pet: Destroying... [id=Mr.seagull]
random_pet.my_pet: Destruction complete after 0s
random_pet.my_pet: Creating...
random_pet.my_pet: Creation complete after 0s [id=Mr.proper.mongoose]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

main.tf

main.tf > resource "random_pet" "my_pet"

```
resource "local_file" "my_pet" {
  filename = var.filename
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = var.length
}
```

variables.tf

variables.tf > variable "length"

```
variable "filename" {

}
variable "length" {

}
```

-

Terraform Tasks - 02

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform apply -auto-approve
var.filename
  Enter a value: Sly

var.length
  Enter a value: 2

local_file.my_pet: Refreshing state... [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]
random_pet.my_pet: Refreshing state... [id=Mr.proper.mongoose]

Terraform used the selected providers to generate the following execution plan. Resource actions a
-/+ destroy and then create replacement

Terraform will perform the following actions:
```

- main.tf > resource "random_pet" "my_pet"

```
resource "local_file" "my_pet" {
  filename = var.filename
  content  = "I love pets!"
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = var.length
}
```

variables.tf > variable "length"

```
variable "filename" {

}
variable "length" {

}
```

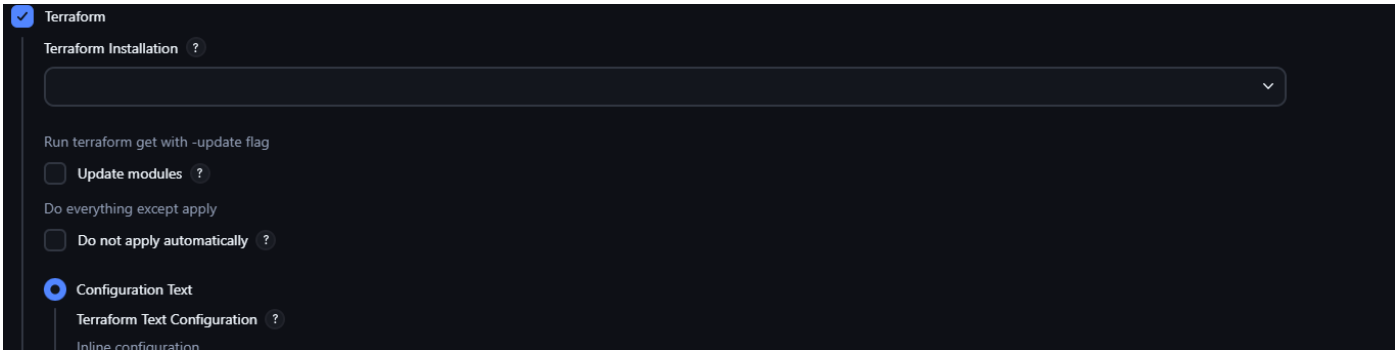
```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform apply -var="filename=pets.txt" -var="length=2"
random_pet.my_pet: Refreshing state... [id=Mr.proper.mongoose]
local_file.my_pet: Refreshing state... [id=7e4db4fbfdbb108bdd04692602bae3e9bd1e1b68]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.my_pet must be replaced
```

- Integrate **Terraform in Jenkins** using the **Terraform plugin**.
  - I have Terraform installed inside my Jenkins container.

- 

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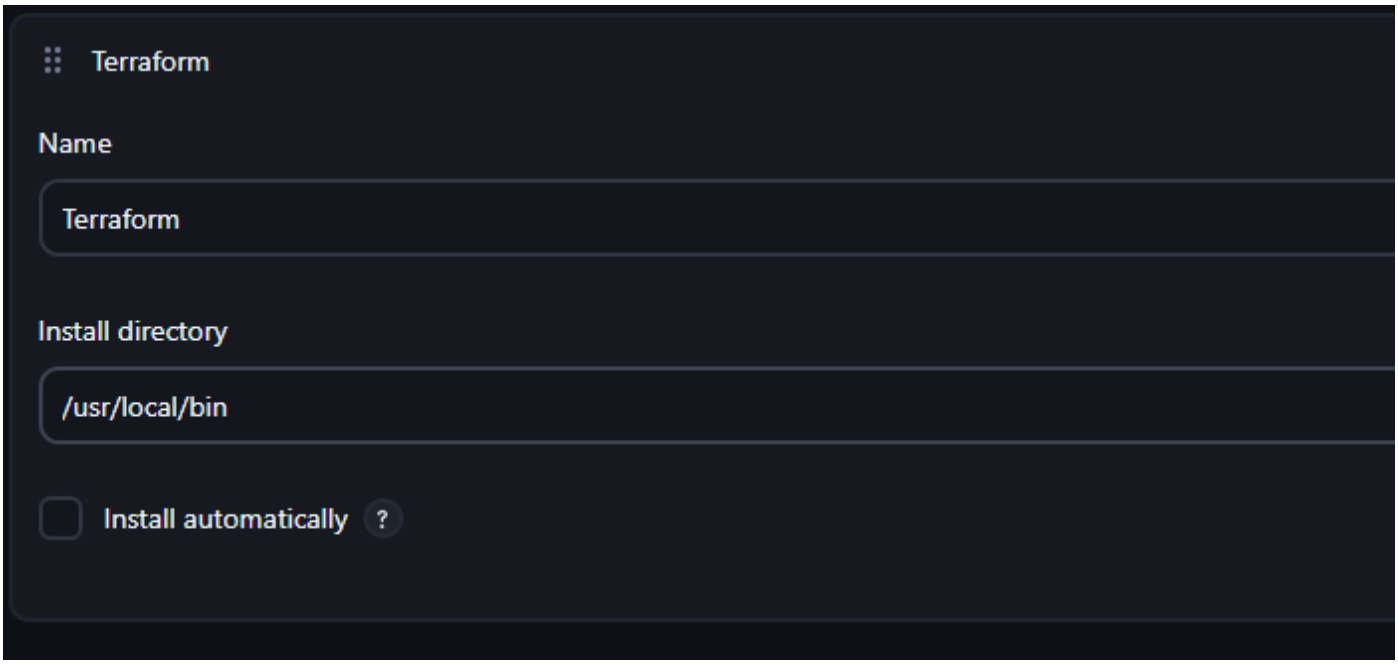
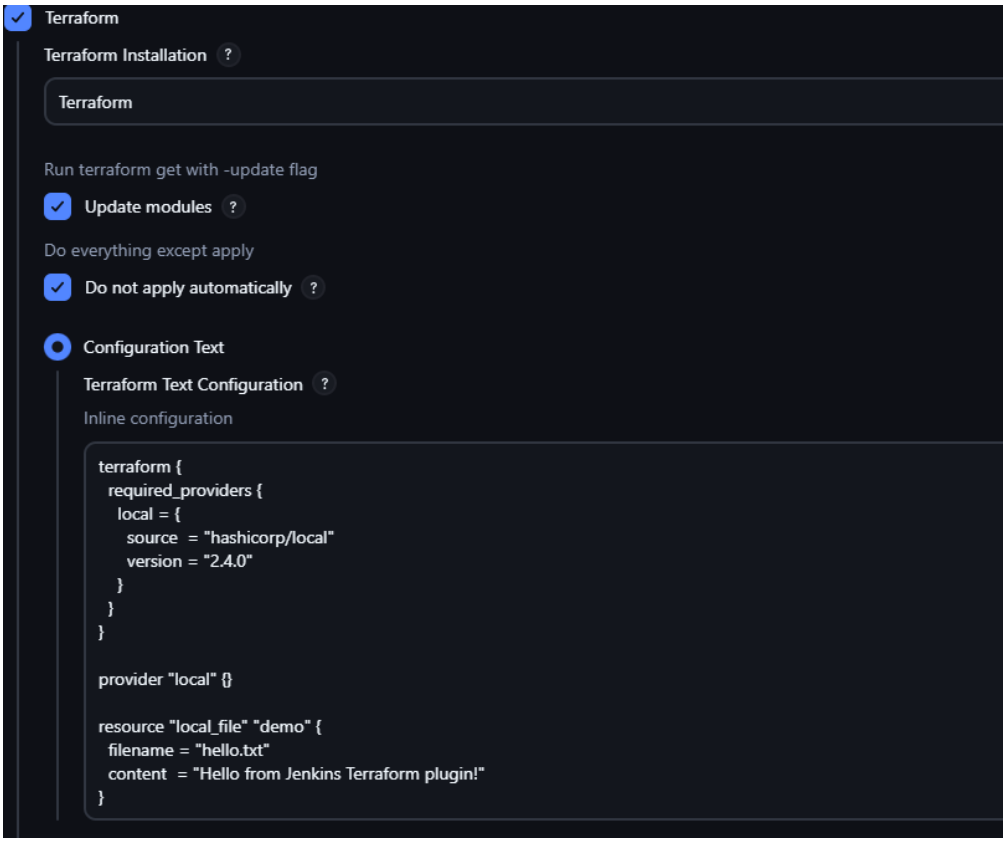


-



Terraform Tasks - 02

```
Fetching changes from the remote Git repository
 > git config remote.origin.url https://github.com/ViqaasTaimoor/Testing.git # timeout=10
Fetching upstream changes from https://github.com/ViqaasTaimoor/Testing.git
 > git --version # timeout=10
 > git --version # 'git version 2.47.3'
 > git fetch --tags --force --progress -- https://github.com/ViqaasTaimoor/Testing.git +refs/heads/*:refs/remotes/
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 2c0643c00c65b1f5fe3b3db2147a48f992122511 (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 2c0643c00c65b1f5fe3b3db2147a48f992122511 # timeout=10
Commit message: "Add local_file resource to create hello.txt"
 > git rev-list --no-walk 2c0643c00c65b1f5fe3b3db2147a48f992122511 # timeout=10
[terraform-plugin] $ /usr/local/bin/terraform get -update
Finished: SUCCESS
```